

MA1513 Weekly Self Practice Set 6 Answer

1. Consider the system of differential equations $\mathbf{y}' = \begin{pmatrix} -2 & 0 \\ 5 & -2 \end{pmatrix} \mathbf{y}$.
- (a) Show that all the eigenvectors of $\mathbf{A} = \begin{pmatrix} -2 & 0 \\ 5 & -2 \end{pmatrix}$ have the form $\begin{pmatrix} 0 \\ r \end{pmatrix}$.
 - (b) Find a generalised eigenvector of $\mathbf{A}$ that corresponds to the eigenvector $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
 - (c) Write down a fundamental set of solutions of the SDE.
 - (d) State the type of equilibrium point of the system and its stability.

Answer

- (a) The only eigenvalue of the triangular matrix $\mathbf{A} = \begin{pmatrix} -2 & 0 \\ 5 & -2 \end{pmatrix}$ is -2 (the diagonal entries).

By solving $(-2\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0} \Rightarrow \begin{pmatrix} 0 & 0 \\ -5 & 0 \end{pmatrix} \mathbf{x} = \mathbf{0}$, we get the solutions $\mathbf{x} = \begin{pmatrix} 0 \\ r \end{pmatrix}$.

So all eigenvalues of $\mathbf{A}$ have the form $\begin{pmatrix} 0 \\ r \end{pmatrix}$.

- (b) To find a generalised eigenvector $\mathbf{u}$ of $\mathbf{A}$ corresponding to $\mathbf{v}$, we solve $(\mathbf{A} + 2\mathbf{I})\mathbf{u} = \mathbf{v} \Rightarrow \begin{pmatrix} 0 & 0 \\ 5 & 0 \end{pmatrix} \mathbf{u} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. We get the solutions $\mathbf{u} = \begin{pmatrix} 1/5 \\ s \end{pmatrix}$.

We let $s = 0$ (can be any other values) to get a specific generalised eigenvector $\mathbf{u} = \begin{pmatrix} 1/5 \\ 0 \end{pmatrix}$.

- (c) From the above, we can get a fundamental set of solutions:

$$\mathbf{y}_1 = \mathbf{v}e^{-2t} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-2t}.$$

$$\mathbf{y}_2 = (t\mathbf{v} + \mathbf{u})e^{-2t} = \begin{pmatrix} 1/5 \\ t \end{pmatrix} e^{-2t}.$$

- (d) Degenerate node; stable.

2. Consider the system of differential equations $\mathbf{y}' = \begin{pmatrix} 3 & 7 \\ -7 & 3 \end{pmatrix} \mathbf{y}$.
- Find the general (real) solution of the system. (You may use MATLAB to find the eigenvalues and eigenvectors of the underlying matrix.)
 - State the type of equilibrium point of the system and its stability.
 - Is the trajectory moving in clockwise or anticlockwise direction? How do you tell?

Answer

- (a) The eigenvalues are the complex conjugate pairs $3 \pm 7i$.

The eigenvector of eigenvalue $3 + 7i$ is given by $\begin{pmatrix} -i \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

Hence the general solution of the system is given by

$$\mathbf{y} = c_1 e^{3t} \begin{pmatrix} \sin 7t \\ \cos 7t \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} -\cos 7t \\ \sin 7t \end{pmatrix}$$

[Alternatively, if we choose eigenvalue $3 - 7i$, the corresponding eigenvector is given by $\begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. In this case, the general solution of the system is given by

$$\mathbf{y} = c_1 e^{3t} \begin{pmatrix} \sin 7t \\ \cos 7t \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} \cos 7t \\ -\sin 7t \end{pmatrix}$$

which is also a correct answer.]

- (b) Spiral; unstable.
- (c) By testing the point $(1, 0)$, the tangent vector of the trajectory at $(1, 0)$ is given by $\begin{pmatrix} 3 & 7 \\ -7 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -7 \end{pmatrix}$, which is pointing downward. So the trajectory is moving clockwise.